

Utkarsh Gupta

Business & Data Analyst | Python & SQL

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SUMMARY

Engineering undergraduate with hands-on experience building data pipelines and quantitative models to analyze real-world financial and operational problems – from power market pricing data to stochastic risk simulations. Comfortable working with messy, ambiguous datasets and translating quantitative findings into clear, decision-ready insights for non-technical stakeholders. Strong foundation in Python-based data analysis, scenario modeling, and structured technical communication. Seeking Business Analyst / Data Analyst roles where analytical rigor and clear communication drive better decisions.

RELEVANT SKILLS

Analysis & Modeling: Data pipeline construction, time-series analysis, Monte Carlo simulation, scenario/financial modeling, statistical analysis

Technical: Python (Pandas, NumPy, Matplotlib), SQL (basic), Git, Excel, Power BI

Communication: Structured technical documentation (Notion, Markdown), stakeholder-ready presentation delivery (AROHAN 2025 conference), translating quantitative findings into plain language

Domain Exposure: Energy markets, DISCOM/utility economics, financial risk modeling, embedded systems

ANALYTICAL EXPERIENCE

Data Analysis & Financial Modeling | *DISCOM Revenue Leakage & Battery Arbitrage Analysis* 2026

- Built a data pipeline in Python analyzing hourly Market Clearing Price (MCP) data from POSOCO and IEX to evaluate battery storage arbitrage potential across Indian states.
- Calculated peak vs. off-peak price spreads to estimate arbitrage margins for battery energy storage systems (BESS), structuring findings into comparative state-level rankings.
- Cleaned and reshaped multi-year time-series datasets using Pandas, resolving structural anomalies in raw government data (e.g., merged administrative regions) before analysis.
- Visualized regional arbitrage opportunity and price-volatility trends using Matplotlib to support data-driven interpretation.

Risk Simulation & Scenario Modeling | *Indore Microgrid Monte Carlo Risk Simulation* 2025 – 2026

- Built a Monte Carlo simulation engine in Python running 2,000 randomized yearly iterations to stress-test a 50-household solar microgrid's financial solvency.
- Modeled stochastic scenarios including monsoon-driven solar generation drops (20–60%) and policy tariff shocks (Rs. 1.00–3.00/kWh) to evaluate downside risk.
- Generated risk heatmaps and savings-distribution histograms using NumPy and Matplotlib to identify cash-flow risk thresholds under adverse conditions.

Systems Modeling & Technical Communication | *Energy Grid Research, AROHAN 2025* Sept – Nov 2025

- Co-authored and presented a techno-economic research paper on decentralized energy storage systems at a national student research conference (IEEE Student Conference).
- Modeled and communicated system-wide financial impact (Rs. 1.91 lakh crore/year baseline AT&C loss exposure) using public grid data and scenario-based assumptions – framed as a theoretical model, not field-validated.

ADDITIONAL TECHNICAL PROJECTS

Stability-Aware Dispatch Model | *Python, NumPy, Matplotlib* 2025 – 2026

- Built an object-oriented dispatch simulation modeling power balance across a hybrid renewable-nuclear industrial microgrid with hydrogen electrolysis load, using rule-based heuristic dispatch logic.
- Tracked grid-import deficits under varying load conditions and visualized system performance trade-offs across scenarios.

Wearable Neck Posture Correction Device | *Arduino Nano, MPU6050* Jan – May 2025

- Developed and calibrated a wearable sensor device with real-time alert logic as part of a college hardware project, debugging hardware-software integration issues through iterative testing.

EDUCATION

Shri G.S. Institute of Technology and Science (SGSITS), RGPV
Bachelor of Technology – Electrical Engineering

Indore, M.P.
2027